IZAK OLTMAN

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RESEARCH INTERESTS

Semiclassical analysis and random perturbations of non-self-adjoint operators.

EMPLOYMENT

Northwestern University *Fall 2024 - Present*
NSF RTG Fellow and Boas Assistant Professor
Postdoc Supervisor: Jared Wunsch

EDUCATION

University of California, Berkeley *Fall 2019 - Spring 2024*
Ph.D. in Mathematics
Advisor: Maciej Zworski

University of Wisconsin, Madison *Fall 2015 - Spring 2019*
Bachelor of Arts – College of Letters and Science
Major in Mathematics with Honors
Certificate in Physics

Budapest Semesters in Mathematics *Spring 2018*
Budapest, Hungary

AWARDS

NSF Graduate Research Fellowship *2021 - 2024*
Dowling Scholarship - UW Madison Math Department *2018*
Dean's List - UW Madison *2015 - 2019*
Vilas Scholarship - UW Madison *2015 - 2018*
AMETEK Scholarship *2015 - 2019*
Green House Summer Research Scholarship - UW Madison *Summer 2016*

PUBLICATIONS AND PREPRINTS

8. *The spectrum of the Scottish flag operator* with Frédéric Klopp (*in preparation*)
7. *A Poisson Formula for the Wave Propagator on Schwarzschild-de Sitter Backgrounds* with Benjamin Pineau (arXiv:2512.16054)
6. *Spectral instability of random Fredholm operators* with Simon Becker and Martin Vogel – in **Journal of Spectral Theory**, 16.1(2026), 335-389 (arXiv:2502.10222)
5. *Absence of small magic angles for disordered tunneling potentials in twisted bilayer graphene* with Simon Becker and Martin Vogel (arXiv:2402.12799) – to appear in the **AMS Contemporary Mathematics** volume in memory of Steve Zelditch
4. *Magic angle (in)stability and mobility edges in disordered Chern insulators* with Simon Becker and Martin Vogel – in **Journal of Physics A: Mathematical and Theoretical**, 58.36(2025) (arXiv:2309.02701)
3. *Particle trajectories for quantum maps* with Yonah Borns-Weil (arXiv:2210.03224) – in **Annales Henri Poincaré**, 25.8(2024), 3699-3738
2. *A probabilistic Weyl-law for perturbed Berezin–Toeplitz operators* – in **Journal of Spectral Theory**, 13.2(2023), 727-754 (arXiv:2207.09599)
1. *An exotic calculus of Berezin–Toeplitz operators* – in **Mathematika**, 2.2(2025) (arXiv:2207.09596)

TALKS/PRESENTATIONS

- *AMS Special Session on Microlocal Analysis and Mathematical Physics*. March 2026. Boston College.
- *Inverse Problems, Spectral, and Scattering Theory Seminar*. March 2026. Purdue University.
- *Analysis and Applied Mathematics Seminar*. February 2026. Bocconi University.
- *Spectral Theory Seminar*. January 2026. UC Berkeley.
- *RTG Dynamics REU Colloquium*. July 2025. Northwestern University.
- *Analysis Seminar*. February 2025. Northwestern University.
- *Probability and Analysis Seminar*. February 2025. University of Michigan, Ann Arbor.
- *Spectral Theory Seminar*. January 2025. UC Berkeley.
- *Joint Mathematics Meeting*. January 2025. Seattle, Washington.
- *International workshop on 2D and moiré materials*. July 2024. Roscoff, France.
- *Harmonic Analysis and Differential Equations Seminar*. February 2024. UC Berkeley.
- *Analysis Seminar*. November 2023. University of Strasbourg.
- *Spectral Theory Seminar*. November 2023. UC Berkeley.
- *Microlocal and Probabilistic Methods in Geometry and Dynamics Summer School (poster presentation)*. July 2023. Paris, France.
- *The 24th Midrasha Mathematicae Random Schrödinger Operators and Random Matrices (poster presentation)*. May 2023. IIAS Jerusalem.
- *Harmonic Analysis and Differential Equations Seminar*. April 2023. UC Berkeley.
- *PDE and Mathematical Physics Seminar*. November 2022. ETH Zurich.
- *Probability and Mathematical Physics Conference (poster presentation)*. July 2022. Helsinki, Finland.
- *Mathematical Challenges in Quantum Mechanics (poster presentation)*. June 2022. Como, Italy.
- *6th Great Lakes Mathematical Physics Conference*. June 2022. Michigan State University.
- *Harmonic Analysis and Differential Equations Seminar*. February 2022. UC Berkeley.

TEACHING

Instructor at Northwestern University

- Math 230-1: multivariable differential calculus (Spring 2026)
- Math 228-1: multivariable differential calculus for Engineering (Fall 2025)
- Math 230-2: multivariable integral calculus (Spring 2025)
- Math 228-1: multivariable differential calculus for Engineering (Fall 2024)

Graduate Student Instructor at UC Berkeley:

- Math 128a: numerical analysis (Spring 2021)
- Math 16a: calculus and analytic geometry (Fall 2020)
- Stat 134: concepts of probability (Summer 2020)
- Math 53: multivariable calculus (Spring 2020)
- Math 54: linear algebra and differential equations (Fall 2019)

CODING LANGUAGES

LaTeX, MATLAB, Python

SERVICE

Organizer and mentor of an REU at Northwestern (2025); mentored Reyna Li and Ian Hollas on modeling quantum trajectories on Kähler manifolds

Officer of Berkeley's Math Graduate Student Association (2020 - 2023)

Private Math Tutor for Berkeley Undergraduates (2019 - 2024)

Organizer of UC Berkeley's Semi-classical Analysis Seminar (2022)

Organizer of UC Berkeley's Harmonic Analysis and Differential Equations Student Seminar (2023 - 2024)